个人简历

Guan Jiaojiao

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Education

2018.09 - 2021.04 **Northwestern Polytechnical University (985)** **Master of Computer Science**

- Rank Certification：Top5%
- Relevant Coursework: Data Mining Techniques (91); Machine Learning Algorithms and Applications (88)；Pattern Classification (88); Computational Biology (88)

2014.09 - 2018.06 **Xin Jiang University (211)** **Bachelor of Computer Science**

- Rank Certification: Top5%
- Relevant Coursework: Advanced mathematics (94); Linear algebra (96)；Probability theory and Mathematical statistics (93); Programming C language (92); C++ (93); Java (96); Operating System (96)

Research Interests

- Graph Network representation learning (GCN, GAT, GAN, Node2vec);
- Survival analysis for Parkinson disease;
- Diseases and phenotype relationship analysis;
- Recommendation System;

Papers

- Jiajie Peng, **Jiaojiao Guan**, and Xuequn Shang. 2019. Predicting Parkinson’s Disease Genes Based on Node2vec and Autoencoder. Frontiers in Genetics 10 (2019), 226. (IF：4.151) (First student author)
- Liu Haijie, **Guan Jiaojiao**, Li He, Bao Zhijie, Wang Qingmei, Luo Xun, Xue Hansheng. 2020. Predicting the Disease Genes of Multiple Sclerosis Based on Network Representation Learning. Frontiers in Genetics. 11 (2020), 328. (IF: 3.517) (First student author)
- Jiajie Peng, **Jiaojiao Guan**, Weiwei Hui and Xuequn Shang. 2020. A novel subnetwork representation learning method for uncovering disease-disease relationships. Methods, Volume 192, 2021, Pages 77-84, ISSN 1046-2023 (IF: 3.81) (First student author)
- Jiajie Peng, Yuxian Wang, **Jiaojiao Guan** and Xuequn Shang. 2020. An end-to-end heterogeneous graph representation learning-based framework for drug-target interaction prediction. Briefings in bioinformatics, bbaa430. 1 Feb. 2021. (IF:8.99) (Second student author)

Patent

Jiajie Peng, Weiwei Hui, Jiaojiao Guan and Xuequn Shang. 2019. Phenotype relationship analysis based on network representation learning.

Work Experience

2021.09 - 2022.06 **City University of Hong Kong** **Research Assistant**

- Analysing next-generation sequencing data
- Designing deep learning algorithm in virus classification

Research Experience

2020.08 - 2021.04 **Predicting gene ontology function of human MicroRNAs based on graph convolutional network.**

MicroRNAs (miRNAs) have been demonstrated to play significant biological roles in many human biological processes. Inferring the functions of miRNAs is an important strategy for understanding disease pathogenesis at the molecular level. In this paper, we propose an integrated model based on graph convolutional network to infer the gene ontology (GO) functions of miRNAs by integrating multiple data sources, including miRNAs-miRNAs intereaction, miRNA-GO interactions, and GO-GO interactions.

- Firstly, we build a global network consisting of three networks by miRNAs similarity and GO term similarity.
- Secondly, we compute the features of miRNAs and GO terms.
- Finally, We constuct an end-to-end model to predict miRNA-GO relationships based on graph convolutional network and multilayer perceptron;

2020.02 - 2020.08 **An end-to-end heterogeneous graph representation learning-based framework for drug-target interaction prediction.**

Accurately identifying potential drug-target interactions (DTIs) is a key step in drug discovery. we develop a novel end-to-end learning-based framework for drug-target interaction prediction based on heterogeneous graph convolutional network. Given a heterogeneous network containing multiple types of biological entities (drug, protein, disease, side-effect), our model learns the low dimensional feature representation of drugs and targets using a graph convolutional network-based model and predicts drug-target interactions based on the learned features.

- First, a heterogeneous network is constructed by combining multiple biological networks, such as drug-drug interaction network, drug-protein interaction network, drug disease association network and so on;
- Second, we propose a novel graph convolutional neural network to obtain the low dimensional representations of drugs and proteins based on the constructed network ;
- Third, we use the inner product method to calculate the score between drugs and proteins based on the low dimensional representations and optimize the model by cross-entropy ;

2019.07 - 2020.02 **A novel subnetwork representation learning method for uncovering disease-disease relationships**

Analyzing disease-disease relationships plays an important role for understanding disease mechanisms and finding alternative uses for a drug. Given a disease and a network, the disease could be represented by a subnetwork constructed by the disease genes involved in the given network, named disease subnetwork. we propose a novel method named SubNet2vec to learn the feature vectors of diseases from its corresponding subnetwork in the biological network.

- Firstly, a bias random walk method is applied to protein-protein interaction (PPI) network for sampling node sequences.
- Secondly, based on a probability model, each node involved in the node sequences is assigned a specific disease subnetwork to generate subnetwork sequences.
- Thirdly, we utilize SkipGram to optimize low-dimensional representations of disease subnetworks based on their context information in the disease subnetwork sequences. Finally, we could apply a supervised model to analyze disease-disease relationships based on gene features and subnetwork features.

2019.02 - 2019.07 **Predicting the Disease Genes of Multiple Sclerosis Based on Network Representation Learning**

Multiple sclerosis (MS) is an autoimmune disease for which it is difficult to find exact disease-related genes. Effectively identifying disease-related genes would contribute to improving the treatment and diagnosis of multiple sclerosis. In this paper, we try to introduce NRL into the task of disease-related gene prediction and propose a novel framework for identifying the disease-related genes multiple sclerosis.

- Capturing the topological structure of the PPI network using NRL-based methods;
- Encoding learned features into low-dimensional space using a stacked autoencoder;
- Training a support vector machine (SVM) classifier to predict disease-related genes;

2018.09 - 2019.02 **Predicting Parkinson's Disease Genes Based on Node2vec and Autoencoder**

Identifying genes associated with Parkinson's disease plays an extremely important role in the diagnosis and treatment of Parkinson's disease. In recent years, based on the guilt-by-association hypothesis, many methods have been proposed to predict disease-related genes, but few of these methods are designed or used for Parkinson's disease gene prediction. In this paper, we propose a novel prediction method for Parkinson's disease gene prediction.

- Extracting features of genes based on Protein Protein Intereaction network using Node2vec;
- Reducing the dimension of gene features using denoise AutoEncoder;
- Predicting Parkinson's disease genes using a machine learning method;

Other Informations

- First-class scholarship of China National (2020); First-class scholarship of China National (2018); First-class scholarship of Xinjiang (2017); Second-class scholarship of Xinjiang (2016) ; Second-class scholarship of Xinjiang (2015)
- In 2019, I obtained the funding of NWPU to sponsor study in Harvard University. (Delayed because of COVID-19)